

A Cloud-Integrated, Quantum-Parameterized Extension of the Encrypted Proximity Recognition (EPR) Framework

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July 13, 2026

Abstract

Proximity-based access control in regulated environments requires cryptographic identity assurance, physical presence enforcement, and tamper-evident audit capability simultaneously, a combination that existing connectionless Bluetooth Low Energy (BLE) deployments do not provide. This paper presents a cloud-integrated extension of the Encrypted Proximity Recognition (EPR) framework incorporating hardware quantum random number generator entropy and a three-party protocol architecture in which physical presence at the access point is enforced as a cryptographic precondition rather than an assumption. The protocol combines a time-based session parameter generated locally at the host and verified through independent server-side re-computation with a quantum-derived session token broadcast over a power-limited BLE channel, such that an adversary must simultaneously defeat two structurally independent physical barriers to advance the protocol. Session credentials are parameterized by values sourced from a hardware quantum random number generator, producing outputs that are physically irreproducible regardless of adversarial capability and eliminating the class of attacks that exploit pseudorandom generator seed predictability in constrained environments. All protocol events are suggested to be recorded in a chained-hash tamper-evident audit log satisfying compliance requirements mandated in healthcare, financial, and government deployment contexts. Statistical evaluation of the quantum random output confirmed properties consistent with ideal uniform independent randomness, and cryptographic analysis of the one-way mixing function demonstrated strong diffusion behavior and zero collisions over a large quantum-derived input sample, consistent with the expected behavior of the underlying hash construction operating under constrained output length. The framework is designed as a general-purpose secure proximity service applicable to facility access control, sensitive equipment authorization, attendance management, and any institutional context where authenticated physical presence is operationally essential.

Keywords: Bluetooth low energy, Proximity Identification, Quantum Random Numbers

1 Introduction

Securing access to sensitive locations and facilities has become a foundational challenge of modern institutional security. Military installations, government, banking, and medical facilities and critical infrastructure sites require identity verification that is cryptographically sound and resistant to spoofing, replay, and remote impersonation. Traditional deployments in these environments combine multi-factor authentication including proximity cards, biometric scanners, and PIN-based entry enforced through purpose-built hardware readers tethered to centralized identity management platforms [4, 6, 10].

The convergence of mobile computing and short-range wireless communication has opened an alternative path. Bluetooth Low Energy (BLE) is now available across smartphones, wearables, and IoT edge devices, offering low power consumption and short-range propagation characteristics naturally suited to proximity verification without requiring specialized readers [2, 7].

Connection-based BLE achieves mutual authentication through pairing and bonding but is fundamentally constrained by hardware-imposed ceilings on concurrent active connections, typically fewer than ten simultaneously paired peripherals on commodity smartphone chipsets, limiting its applicability to multi-user environments [3]. The pairing process itself increases connection setup time, and once a paired connection is established it can introduce unwanted data transfer and additional attack surface [18]. Connectionless BLE advertising circumvents this bottleneck entirely, but traditional beacon implementations broadcast static or weakly protected identifiers with no cryptographic assurance of authenticity, leaving them vulnerable to cloning, replay injection, and Sybil attacks [11, 13].

Location awareness has become an equally important dimension of access security. Outdoor deployments leverage geofencing to flag access attempts originating outside expected geographic regions [8]. Indoors, where satellite signals are attenuated or absent, WiFi fingerprinting and BLE RSSI-based triangulation are used to approximate device position, though WiFi positioning requires dense access point infrastructure and RSSI estimation is sensitive to multipath interference and hardware variation [12, 15]. Location sensing alone is insufficient. What regulated environments require is the fusion of spatial evidence with cryptographic identity proofs, producing a verifiable claim that an authorized entity is genuinely present at a specific location at a specific time.

The quality of randomness underlying session credential generation is a dimension that receives insufficient attention in proximity authentication. Most deployed systems rely on pseudorandom number generators seeded from software entropy pools which, while statistically adequate under benign conditions, are potentially predictable in constrained environments and vulnerable where an attacker may observe or influence the entropy state [19, 21]. Quantum random number generators address this at a foundational level by deriving entropy from provably non-deterministic physical processes such as photon path superposition, vacuum fluctuations, or radioactive decay, producing randomness that is not merely computationally unpredictable but physically irreproducible regardless of prior knowledge of the generator state [1, 9, 14]. Despite this well-established advantage, QRNG integration into proximity authentication protocols remains largely unexplored.

Cloud-native architecture addresses a separate but equally critical gap. Regulated industries operate under compliance obligations such as HIPAA in healthcare, PCI-DSS in financial infrastructure, and FISMA in government systems, all of which mandate tamper-evident and centrally accessible audit logs of every access event including timestamps, identities, session parameters, and outcomes [17]. A device-local proximity system cannot satisfy these requirements, nor can it support real-time credential revocation, cross-site identity recognition, or intelligent anomaly detection over behavioral patterns. Machine learning based fraud detection operating over centralized logs enables the identification of coordinated attacks, credential farming, and anomalous access patterns that classical rate-limiting alone cannot detect [16, 20].

Our prior work introduced the Encrypted Proximity Recognition (EPR) framework, a connectionless BLE protocol achieving cryptographic device authentication through one-way key mixing without pairing overhead, validated at 100% identification success across 50 or more concurrent devices [5]. That framework operated in a device-local model with software-generated session parameters and no centralized logging or fraud management. This paper presents a cloud-integrated extension of EPR incorporating QRNG-sourced session entropy and a three-party protocol architecture in which physical proximity is enforced as a hard cryptographic requirement. The user must physically receive a controlled short-range BLE broadcast from the host verifier to obtain a session token, which combined with their long-term key is submitted to a cloud server for quantum-parameterized credential generation and verification. The framework is designed as a general-purpose secure proximity service applicable to facility access control, sensitive equipment authorization, attendance management, and any context where authenticated physical presence is essential and critical.

The reference implementation of the proposed framework is available as open-source code at <https://github.com/dwaidatta/encrypted-proximity-recognition-epr-research>.

2 Materials and Methods

2.1 Prior EPR Framework

The original EPR framework established cryptographic proximity authentication over connectionless BLE advertising without requiring pairing, bonding, or any persistent connection between devices [5]. Each enrolled device i was pre-distributed a long-term secret key K_i derived from device-specific attributes through a keyed derivation function. At each time epoch t , the host verifier independently computed a session parameter $\mathcal{P}_t$ as a keyed hash over the current epoch and its own locally stored secret, formatted to a fixed alphanumeric length l chosen by the deployment operator. This construction followed the TOTP standard, ensuring $\mathcal{P}_t$ was fresh per epoch, deterministic, and identically computable by the server for independent verification without requiring the host to transmit it. $\mathcal{P}_t$ was displayed locally at the physical access point for the user to observe on-site.

Upon observing $\mathcal{P}_t$, each enrolled device computed a broadcast value $\mathcal{F}_i$ by applying a one-way mixing function over K_i and $\mathcal{P}_t$ through cyclic XOR over the key and parameter bytes, followed by a bitwise permutation step for diffusion, and a cryptographic hash producing a fixed-length output. The device then advertised $\mathcal{F}_i$ continuously over BLE as a non-connectable undirected advertisement payload. The host verifier, operating in scanning mode, precomputed the expected $\mathcal{F}_i$ for every enrolled device i using the same mixing function and compared incoming BLE advertisement payloads against this precomputed set. A match on any $\mathcal{F}_i$ constituted identification of device i as physically present within BLE range. Because $\mathcal{F}_i$ is a one-way function of K_i and $\mathcal{P}_t$, an observer of the BLE channel cannot recover K_i from $\mathcal{F}_i$, and because $\mathcal{P}_t$ rotates every T_{valid} minutes, a captured $\mathcal{F}_i$ is invalid in any subsequent epoch. The framework achieved 100% identification success across 50 or more concurrently advertising devices on commodity smartphone hardware, demonstrating that connectionless BLE advertising is a viable medium for scalable cryptographic proximity verification without dedicated infrastructure. Figure 1 depicts the original EPR setup.

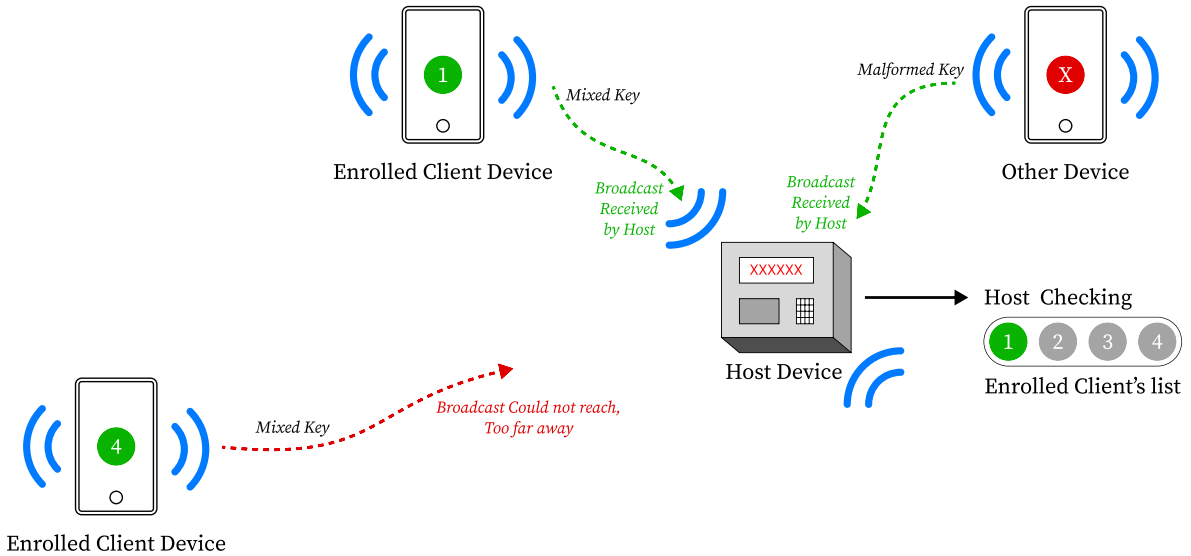


Figure 1: Original Encrypted Proximity Recognition (EPR) Framework using Connectionless Bluetooth Low Energy (BLE). Only one-way broadcast between Client and Host exists.

The device-local model, however, carries structural limitations that constrain its applicability to regulated and high-assurance environments. Software-derived entropy sources for session parameter generation, while statistically adequate under benign conditions, remain deterministic functions of an internal seed state and are potentially predictable under adversarial observation or entropy pool manipulation. The absence of centralized logging renders the system unable to satisfy tamper-evident audit trail requirements mandated by compliance frameworks such as HIPAA, PCI-DSS, and FISMA. Real-time credential revocation, cross-site identity recognition, and behavioral anomaly detection over historical access patterns are similarly infeasible in a purely device-local deployment. The present work extends the EPR framework to address these limitations by introducing a three-party cloud-integrated architecture with quantum parameterization.

2.2 Proposed System Model

The proposed framework involves three logical entities. The user device is a commodity smartphone or similar BLE-capable device operated by an individual seeking access. The host is a locally deployed device at the access point responsible for session parameter generation and physical BLE monitoring. The cloud server acts as the central authority managing enrollment, session validation, quantum-parameterized credential generation, and audit logging. Communication between the user and server and between the host and server occurs over authenticated encrypted channels. The BLE channel between the user and host is inherently unencrypted and is treated as a fully observable medium accessible to any passive or active adversary within radio range as per BLE specifications.

Each user is assigned a unique identifier ID_i and a long-term secret key K_i at enrollment. Since K_i is never transmitted over BLE and travels exclusively over server-authenticated TLS channels, it is not subject to BLE payload constraints and may be of arbitrary length, functioning effectively as a high-entropy access token. Each host h is similarly registered with the server under identifier ID_h . The server maintains a protected database of tuples (ID_i, K_i) for all enrolled users and a registry of all enrolled hosts.

2.3 Registration and Key Distribution

Registration is a one-time process performed through a secure out-of-band channel. The server generates K_i as:

$$K_i = \text{HKDF}(\text{seed}_i, \text{salt}, \text{context}) \quad (1)$$

where seed_i is derived from user-specific attributes, salt is a system-wide constant, and HKDF denotes the HMAC-based key derivation function. The resulting key is delivered to the user device through a secure authenticated channel and stored in device-protected secure storage. The server retains (ID_i, K_i) and marks the enrollment record as active. Key revocation is performed server-side by marking the record inactive, taking effect immediately on the next access request without requiring physical access to the user device.

2.4 Protocol Specification

The protocol operates across eight phases. Let $\mathcal{U}$ denote the set of all enrolled users, $\mathcal{H}$ denote the set of all registered hosts, and $\mathcal{S}$ denote the cloud server. Each session is uniquely identified by a session descriptor σ bound to a specific user-host pair (ID_i, ID_h) and a time epoch t . The session descriptor σ is generated by $\mathcal{S}$ at the start of each access attempt and used to correlate all events and log entries belonging to that attempt. Figure 2 gives an overview of the proposed protocol and its different phases.

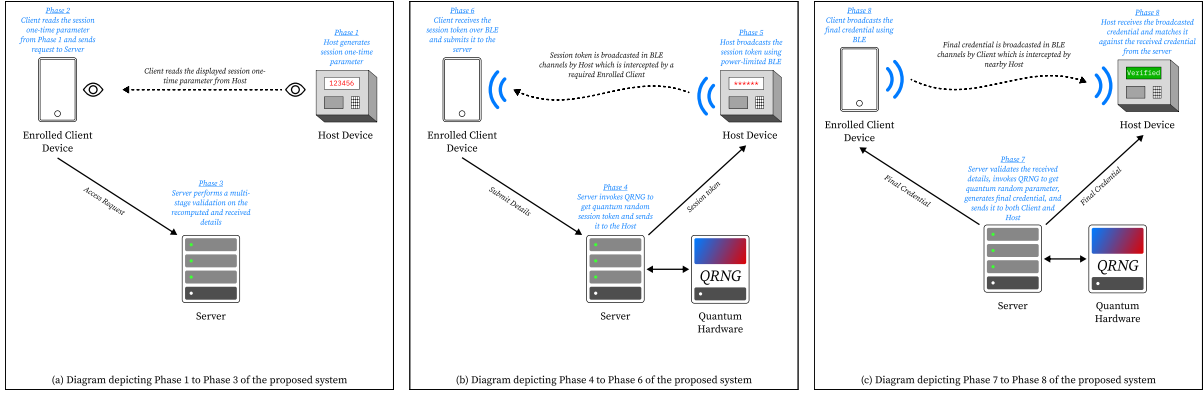


Figure 2: Diagram explaining the different phases of the new proposed protocol.

2.4.1 Phase 1 – Session Parameter Generation

At the start of each time epoch t , the host ID_h independently generates a session one-time parameter $\mathcal{P}_t$. This parameter is valid for a fixed window T_{valid} , recommended to be between two and five minutes depending on deployment security requirements, after which it rotates automatically. The parameter is computed locally on the host using a keyed hash construction over the current time epoch, ensuring it is both deterministic and reproducible by $\mathcal{S}$ for independent verification without requiring the host to transmit it. The host's computation uses its own locally stored secret K_h so that no external party can predict $\mathcal{P}_t$ without knowledge of K_h . The resulting parameter is displayed locally at the physical access point for the user to observe on-site.

Formally, $\mathcal{P}_t$ is computed as:

$$\mathcal{P}_t = \text{Trunc}_d(\text{HMAC-SHA256}(K_h, \lfloor \tau/T_{\text{valid}} \rfloor)) \quad (2)$$

where τ is the current Unix timestamp, $\lfloor \cdot \rfloor$ denotes the floor function, and Trunc_d extracts d output characters, with d chosen by the deployment operator according to usability and security requirements. This construction follows the Time-based One-Time Password (TOTP) standard, ensuring that $\mathcal{P}_t$ is fresh per epoch and identical in value whether computed at the host or independently recomputed at $\mathcal{S}$.

2.4.2 Phase 2 – User Access Request

The user ID_i , upon physically observing $\mathcal{P}_t$ displayed or shared at the host ID_h location, initiates an access request to $\mathcal{S}$. The request carries the user's identity ID_i , their long-term key K_i , the observed session parameter $\mathcal{P}_t$, and the identifier of the host ID_h they are requesting access through. The mapping between a user's physical location and the corresponding ID_h is resolved by the user application through any suitable mechanism such as QR code scanning, NFC, or a preconfigured location registry.

The long-term key K_i serves as a bearer authentication token for the request to $\mathcal{S}$. Any party who intercepts $\mathcal{P}_t$ over the air or by observation is therefore unable to construct a valid request without simultaneously possessing a registered K_i , which never appears on any wireless channel. The request tuple is expressed as:

$$\mathcal{R} = (ID_i, K_i, \mathcal{P}_t, ID_h) \quad (3)$$

2.4.3 Phase 3 – Server-Side Verification

Upon receiving $\mathcal{R}$, the server $\mathcal{S}$ performs a multi-stage verification before proceeding. First, $\mathcal{S}$ independently recomputes the expected session parameter $\hat{\mathcal{P}}_t$ for host ID_h at the current

epoch t using equation (2), since $\mathcal{S}$ holds K_h for every registered host in $\mathcal{H}$. It then compares this against the user-submitted $\mathcal{P}_t$. This step constitutes a server-mediated proof of physical co-location: the user ID_i cannot know the correct $\mathcal{P}_t$ without being physically present at ID_h , since the parameter is displayed locally and rotates every T_{valid} minutes. A user attempting to request access remotely, without physical presence, will be unable to supply the correct current $\mathcal{P}_t$, unless explicitly shared.

Second, $\mathcal{S}$ verifies that K_i corresponds to an active, non-revoked enrollment record for ID_i in $\mathcal{U}$. Third, $\mathcal{S}$ evaluates all applicable access control conditions including rate limit thresholds, revocation status, geographic consistency relative to the claimed ID_h , and any active fraud flags raised by prior sessions. The core verification is expressed as:

$$\mathcal{P}_t \stackrel{?}{=} \hat{\mathcal{P}}_t \quad (4)$$

Failure at any condition results in immediate rejection, a fraud log entry against σ , and enforcement action determined by the deployment policy.

2.4.4 Phase 4 – Quantum Session Token Generation

Upon successful verification, $\mathcal{S}$ invokes the quantum random number generator to produce a session token $\mathcal{Q}_t$. Unlike pseudorandom number generators, which derive output deterministically from an internal seed state and are reproducible given sufficient knowledge of the generator’s prior state, a hardware quantum random number generator derives entropy from a physical quantum process that is fundamentally indeterminate. Processes such as photon path selection at a beamsplitter, vacuum electromagnetic fluctuations measured by homodyne detection, or radioactive decay event timing produce outcomes that are not merely computationally unpredictable but irreducible to any prior cause. Even a complete description of the system’s physical state before the measurement does not determine the outcome. As a result, $\mathcal{Q}_t$ cannot be predicted, precomputed, or reproduced by any party including $\mathcal{S}$ itself.

The generation is expressed as:

$$\mathcal{Q}_t = \text{QRNG}(\eta) \quad (5)$$

where η denotes the required entropy length in bits, chosen to meet the security requirements of the deployment. $\mathcal{S}$ records $\mathcal{Q}_t$ against session σ and transmits it to host ID_h over the authenticated server-to-host channel.

2.4.5 Phase 5 – Host BLE Broadcast

Upon receiving $\mathcal{Q}_t$, the host ID_h begins advertising it over BLE at a controlled transmission power level P_{tx} , calibrated to restrict the effective broadcast range to the physical boundary of the intended access area. This power control is a deliberate physical security mechanism ensuring that $\mathcal{Q}_t$ is receivable only by a device genuinely within the access space, preventing remote interception from outside the controlled area. The value $\mathcal{Q}_t$ is broadcast without further cryptographic transformation, which is architecturally justified because $\mathcal{Q}_t$ carries no long-term enrolled secret. It is single-use, session-bound, and operationally inert to any party who does not also possess a valid K_i registered with $\mathcal{S}$. The server $\mathcal{S}$ simultaneously instructs the user application to begin BLE scanning for session σ .

2.4.6 Phase 6 – Physical Reception and Submission

The user device scans BLE advertising channels and receives $\mathcal{Q}_t$ from the host ID_h broadcast. The requirement that the user physically receive $\mathcal{Q}_t$ over a power-limited BLE channel enforces proximity as a hard cryptographic precondition. Unlike the co-location proof in Phase 3, which relies on the user’s knowledge of $\mathcal{P}_t$, this phase requires the user’s device to be materially within

range of the host hardware. An adversary stationed outside the physical access area cannot obtain $\mathcal{Q}_t$ regardless of their knowledge of other protocol parameters.

The user application submits to $\mathcal{S}$:

$$\mathcal{M} = (\text{ID}_i, K_i, \mathcal{Q}_t) \quad (6)$$

The server $\mathcal{S}$ verifies that $\mathcal{Q}_t$ in $\mathcal{M}$ matches the value issued to ID_h for session σ , and that K_i remains valid. A confirmed match jointly establishes two independent proofs: possession of a valid enrolled K_i , and physical presence within the broadcast range of ID_h . Both conditions must be satisfied simultaneously. Compromising one without the other is insufficient for an adversary to advance the protocol.

2.4.7 Phase 7 – Final Credential Generation

Upon successful validation of $\mathcal{M}$, $\mathcal{S}$ invokes the QRNG a second time to obtain an independent quantum entropy parameter θ_t :

$$\theta_t = \text{QRNG}(\eta) \quad (7)$$

$\mathcal{S}$ then computes the final access credential $\mathcal{F}_i$ by applying a one-way mixing function $g(\cdot)$ over the user's long-term key K_i , the quantum session token $\mathcal{Q}_t$, the user identifier ID_i , and the entropy parameter θ_t . The mixing function combines these inputs through cyclic XOR, bitwise permutation for diffusion, and a cryptographic hash, producing an output from which no input component can be recovered due to the preimage resistance of the underlying hash construction. The credential $\mathcal{F}_i$ is therefore session-specific, non-reproducible across sessions, and infeasible to invert.

$$\mathcal{F}_i = g(K_i, \mathcal{Q}_t, \text{ID}_i, \theta_t) \quad (8)$$

where:

$$\mathcal{F}_i = \text{Select}_{s,l}(\text{SHA-256}(\text{Temper}(K_i \oplus_{\text{cyc}} \mathcal{Q}_t) \parallel \text{ID}_i \parallel \theta_t)) \quad (9)$$

Here $\oplus_{\text{cyc}}$ denotes cyclic XOR applied over inputs of differing lengths, $\text{Temper}(\cdot)$ denotes a parameterized sequence of bitwise shifts and rotations applied for diffusion, $\parallel$ denotes concatenation, and $\text{Select}_{s,l}$ extracts l characters beginning at offset s within the hash output, where both s and l are deployment-defined parameters chosen by the operator. $\mathcal{S}$ transmits $\mathcal{F}_i$ to both ID_i and ID_h over their respective authenticated channels.

2.4.8 Phase 8 – BLE Role Reversal and Access Grant

In this final phase, the BLE roles of the user and host are reversed relative to Phase 5. The user device, having received $\mathcal{F}_i$ from $\mathcal{S}$, begins advertising it over BLE. The host ID_h , holding the same $\mathcal{F}_i$ received from $\mathcal{S}$, switches from advertising to scanning mode and matches incoming BLE advertisements against the expected credential. On a confirmed match, ID_h grants physical access and logs the outcome against σ on $\mathcal{S}$.

Across the entirety of the protocol, the long-term key K_i is never present on the BLE channel. The only values broadcast over BLE are $\mathcal{Q}_t$ in Phase 5 and $\mathcal{F}_i$ in Phase 8. Both are single-use and session-bound. $\mathcal{Q}_t$ is inert without K_i , and $\mathcal{F}_i$ is a one-way function of enrolled secrets that cannot be reversed to expose any input. Passive interception of either value by an adversary on the BLE channel yields no recoverable secret and no reusable credential.

2.5 QRNG Integration

The framework employs a hardware quantum random number generator at two points in the protocol, specifically in Phase 4 for quantum session token generation and in Phase 7 for the entropy parameter θ_t . Quantum random number generators derive entropy from physical processes governed by quantum mechanical principles, such as vacuum electromagnetic fluctuations

measured through homodyne detection, photon path selection at a beamsplitter, or radioactive decay event timing. These processes are fundamentally indeterminate in the sense that their outcomes cannot be predicted or reproduced from any prior description of the system’s physical state, distinguishing them categorically from pseudorandom number generators whose outputs are deterministic functions of an internal seed.

In the proposed framework, the QRNG is queried with a configurable entropy length η in bits, sufficient to meet the security requirements of the deployment. The raw quantum-derived output is used directly as Q_t in Phase 4 and as the entropy parameter θ_t in Phase 7. Because both values are independently sampled from a non-deterministic physical source, no party including the server itself can predict or precompute either value prior to the moment of generation. Deployments in environments where a dedicated hardware QRNG is unavailable may interface with a remote quantum entropy service through an authenticated API channel, provided the communication channel between the server and the entropy source is secured against interception and substitution. Sessions generated using a non-quantum entropy fallback should be flagged distinctly in the audit log for compliance and forensic purposes.

2.6 Cloud Architecture and Audit Logging

The server $\mathcal{S}$ maintains four logical data stores. The enrollment store holds the tuple $(ID_i, K_i, \text{status})$ for every enrolled user and an equivalent record for every registered host, where status reflects whether the record is active, revoked, or suspended. The session store tracks all in-progress and recently completed sessions, each identified by the session descriptor σ and carrying the current phase state, the issued Q_t , the expiry timestamp, and the identifiers of the participating user and host. The access log is a tamper-evident append-only record of every protocol event.

Every entry in the access log carries a full event record including the session descriptor σ , a UTC timestamp, the identities ID_i and ID_h , the protocol phase at which the event was recorded, the outcome of that phase, any anomaly flags raised, and whether quantum or fallback entropy was used for the session. Tamper evidence is maintained through a chained hash construction in which each log entry L_n incorporates a cryptographic hash of the preceding entry L_{n-1} :

$$L_n.\text{hash} = \text{SHA-256}(L_{n-1}.\text{hash} \parallel L_n.\text{data}) \quad (10)$$

Any retrospective modification of a historical entry invalidates the hash chain from that point forward, making tampering detectable upon verification. This design satisfies the audit trail integrity requirements of regulated deployment environments including those governed by HIPAA, PCI-DSS, and FISMA.

3 Results and Discussion

3.1 Quantum Randomness Evaluation

The statistical properties of a quantum random number generator (taken from Australian National University Quantum Numbers – ANU QRNG) output were evaluated against a cryptographically secure pseudorandom number generator based on operating system entropy and a seeded software pseudorandom number generator. A sample of 102,400 bytes was collected from the ANU QRNG across 100 API requests, and an equivalent volume of data was generated from each software source for direct comparison. Four standard statistical tests were applied: bit frequency proportion, Shannon entropy estimation, serial correlation between successive bytes, and runs ratio across the bit stream. Table 1 presents the results across all three sources.

All three sources demonstrated statistical properties closely consistent with ideal uniform independent randomness across every metric. Ones proportions remained within 0.1% of the ideal value of 0.5, Shannon entropy estimates approached the theoretical maximum of 8.0 bits

Table 1: Statistical Randomness Test Results (102,400 bytes per source)

Metric	QRNG	CSPRNG	PRNG
Ones Proportion	0.500582	0.499243	0.500164
Deviation from Ideal	0.000582	0.000757	0.000164
Shannon Entropy (bits)	7.998066	7.998122	7.998507
Serial Correlation	-0.001427	-0.002604	-0.003083
Runs Ratio	0.500497	0.499226	0.499692

per byte, serial correlation values were negligible in all cases, and runs ratios closely approximated 0.5 as expected of an unbiased bit stream. The statistical indistinguishability of QRNG output from CSPRNG and PRNG output is an expected result and does not diminish the security argument for QRNG use. The critical distinction is not statistical profile but physical reproducibility. A software PRNG is a deterministic function of its seed state and can be reconstructed by any party with sufficient knowledge of that state. A CSPRNG draws from operating system entropy pools that, while practically unpredictable, remain computationally derived. A hardware QRNG derives each output from a quantum physical measurement whose outcome is fundamentally indeterminate and irreducible to any prior cause, making session credentials generated from QRNG entropy physically irreproducible regardless of adversarial capability. This property is the security motivation for QRNG integration in the proposed framework rather than any measurable advantage in standard statistical test outcomes.

3.2 Mixing Function Analysis

The cryptographic mixing function employed in Phase 7 of the protocol was evaluated for two properties: diffusion strength through an avalanche effect analysis, and collision resistance over a large sample. Test inputs were generated using 10 API requests to the ANU QRNG, yielding 10,240 bytes of quantum-derived input material. The function was evaluated with a fixed entropy parameter θ held constant across all runs to isolate the effect of input variation on output behavior, consistent with standard avalanche testing methodology. For the avalanche effect evaluation, a single bit was flipped at each of the first 64-bit positions of the user key input across 500 test samples, and the percentage of output bits that changed in response to each single-bit flip was recorded and averaged. Table 2 presents the results.

Table 2: Mixing Function Analysis Results

Metric	Result
Average Bit Change (%)	40.16
Minimum Bit Change (%)	35.91
Maximum Bit Change (%)	43.6
Collisions per 1000 inputs	0

The average output bit change of 40.16% is below the theoretical ideal of 50% expected of a fully diffusive cryptographic construction. This deviation arises directly from the output truncation applied in the Select operation of equation (9), which extracts an 8-character window from a 256-bit SHA-256 digest. Bit flip effects that propagate into regions of the digest outside the selected window are not observable in the output, reducing the measured avalanche percentage relative to what the full digest would exhibit. This is an inherent tradeoff of constrained output length and does not represent a weakness in the underlying hash construction, whose full 256-bit output exhibits the avalanche property by design. The 8-character output length is a deliberate design parameter retained from the original EPR framework to satisfy

BLE advertising payload constraints. Collision resistance was evaluated over 1000 distinct input combinations drawn from quantum-derived material, and no collisions were observed, yielding a collision rate of zero over the tested sample. While this does not constitute a formal cryptographic proof of collision resistance, it is consistent with the expected behavior of a SHA-256 based construction for which no practical collision attack is known.

3.3 Security Discussion

The security of the proposed protocol rests on the simultaneous enforcement of two independent physical and cryptographic barriers against unauthorized access, neither of which can be bypassed through network access alone. The first barrier is established in Phase 3 through server-mediated verification of the session parameter $\mathcal{P}_t$, generated locally at the host using a keyed TOTP construction, displayed physically at the access point, and verified by the server through independent recomputation. A party attempting access without physical presence at ID_h cannot supply the correct $\mathcal{P}_t$ since it rotates every T_{valid} minutes and is not transmitted over any network channel accessible to a remote adversary. The second barrier is established in Phase 6 through the requirement that the user physically receive $\mathcal{Q}_t$ over the host’s power-limited BLE broadcast, calibrated to the physical boundary of the access area. A device outside that boundary cannot receive $\mathcal{Q}_t$ regardless of knowledge of any other protocol parameter, and the user must submit it alongside K_i to the server, confirming both physical presence and enrollment simultaneously.

These two barriers are structurally independent. An adversary who obtains $\mathcal{P}_t$ by observation but lacks a registered K_i is rejected in Phase 3. An adversary with a stolen K_i but without physical presence cannot receive $\mathcal{Q}_t$ and is rejected in Phase 6. Only a party simultaneously enrolled and physically present can advance through both barriers. The long-term key K_i is never present on the BLE channel at any point in the protocol. The values broadcast over BLE, namely $\mathcal{Q}_t$ and $\mathcal{F}_i$, are both single-use and session-bound. $\mathcal{Q}_t$ carries no enrolled secret and is inert without a valid K_i , while $\mathcal{F}_i$ is the output of a one-way mixing function over enrolled secrets that cannot be inverted to recover any input component. Replay attacks are prevented by session-specific binding of all credentials to QRNG-derived values that are consumed and invalidated upon use, eliminating the class of attacks that exploit PRNG seed predictability in constrained or embedded environments.

3.4 Limitations and Future Work

The mixing function evaluation was conducted under controlled conditions using synthetically generated inputs rather than live protocol sessions, which is standard practice for cryptographic function analysis but does not capture the full distribution of inputs arising in a deployed institutional environment. Large-scale real-world deployment evaluation remains a direction for future work. The framework’s reliance on an external QRNG service introduces a dependency on network availability and API service continuity, and deployments in air-gapped or connectivity-constrained environments would require a locally housed hardware QRNG device. BLE signal propagation introduces inherent spatial ambiguity sensitive to physical obstacles, device hardware variation, and environmental interference, and integration of RSSI-based distance estimation or complementary technologies such as UWB could improve spatial precision in future deployments. Finally, the framework authenticates devices rather than individuals in its base configuration, and on-device biometric verification prior to Phase 2 submission would extend it to multi-factor authentication combining device possession, user presence, and physical proximity without transmitting biometric data beyond the device.

4 Conclusion

This paper presented a cloud-integrated extension of the Encrypted Proximity Recognition framework incorporating hardware quantum random number generator entropy and a three-party protocol architecture in which physical presence at the access point is enforced as a cryptographic requirement rather than an assumption. The protocol operates across eight phases, combining a TOTP-based session parameter verified through server-side recomputation with a quantum-derived session token broadcast over a power-limited BLE channel, such that an adversary must simultaneously defeat two structurally independent physical barriers to advance the protocol. Session credentials are parameterized by QRNG-derived values that are physically non-reproducible, eliminating the class of attacks that exploit pseudorandom number generator seed predictability in constrained or embedded environments. All protocol events are recorded in a chained-hash tamper-evident audit log satisfying the integrity requirements of regulated deployment contexts including HIPAA, PCI-DSS, and FISMA.

Statistical evaluation of the QRNG output confirmed properties consistent with ideal uniform independent randomness across frequency, entropy, serial correlation, and runs metrics, with the critical security distinction from software-based sources residing in physical irreproducibility rather than statistical profile. Cryptographic analysis of the mixing function demonstrated an average avalanche effect of 40.16% across 500 test samples and zero collisions over 1000 distinct input combinations, consistent with the expected behavior of a SHA-256 based construction operating under constrained output length. The framework is designed as a general-purpose secure proximity service applicable to facility access control, sensitive equipment authorization, and any institutional context where authenticated physical presence is operationally essential. Future work includes large-scale real-world deployment evaluation, integration of RSSI-based distance estimation for improved spatial precision, and extension to multi-factor authentication combining device possession, physical proximity, and on-device biometric verification.

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